

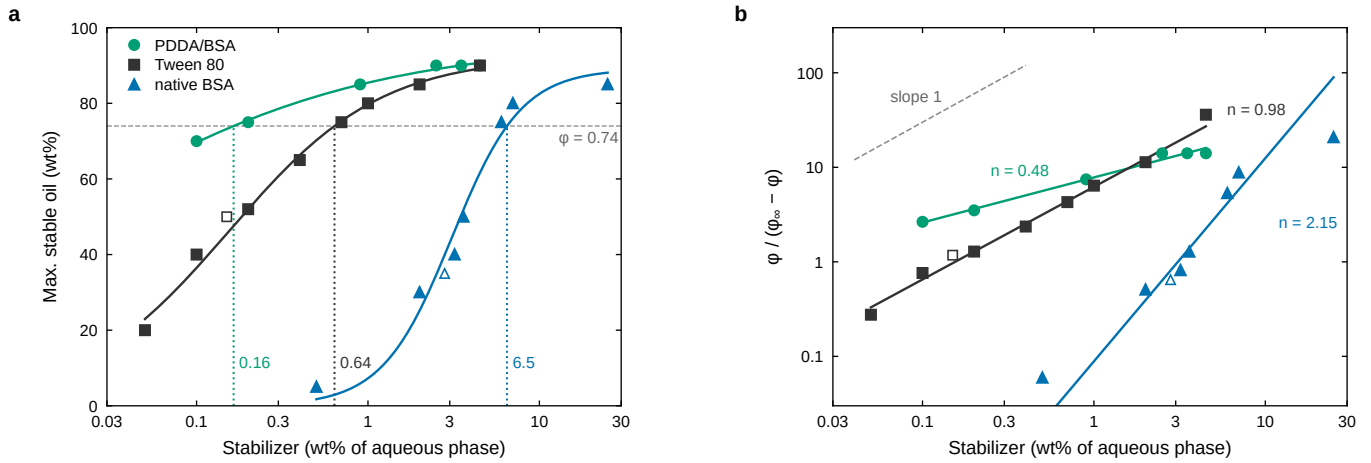


Figure | Maximum stable oil fraction versus stabilizer concentration, fitted with a coverage-based Hill (log-logistic) model. a, Maximum stable oil fraction as a function of stabilizer concentration (wt% of the aqueous phase) for the PDDA/BSA coacervate (green circles), Tween 80 (grey squares) and native BSA (blue triangles). Solid lines are least-squares fits of $\phi_{\max} = \phi_\infty \cdot c^n / (c_{50}^n + c^n)$, drawn over the measured concentration range of each series. The dashed horizontal line marks $\phi = 0.74$ (close packing of monodisperse spheres; above it the emulsions are high-internal-phase emulsions); dotted vertical lines and numbers give the fitted concentration c^* at which each curve reaches 0.74. **b,** Linearized form of the same data, $\phi/(\phi_\infty - \phi)$ versus c on log-log axes, using the fitted ϕ_∞ of each system; in this representation the model is a straight line of slope n . The grey dashed segment has slope 1, the value expected when the stabilizer only has to cover the oil–water interface of droplets of fixed size. Filled symbols were used in the fits; open symbols are shown but were not fitted. Points are as plotted in the original figure, which shows no error bars (6, 8 and 7 fitted points for PDDA/BSA, Tween 80 and native BSA). Values were digitized from the original figure and are approximate.

System	ϕ_∞ (wt%)	c_{50} (wt%)	n	RMSE (wt%)	c^* at $\phi = 0.74$ (wt%)
PDDA/BSA	96 ± 3	0.0134 ± 0.003	0.48 ± 0.10	0.6	0.16
Tween 80	92 ± 3	0.155 ± 0.01	0.98 ± 0.08	1.7	0.64
native BSA	89 ± 6	3.09 ± 0.3	2.15 ± 0.45	4.0	6.5

Fit parameters ($\pm$ standard error from the fit covariance).

Fitting approach and interpretation

What we used. Each series was fitted with $\phi_{\max} = \phi_\infty \cdot c^n / (c_{50}^n + c^n)$, using non-linear least squares on the untransformed oil fractions (scipy *curve_fit*). There are three parameters: ϕ_∞ , the plateau oil fraction reached at high stabilizer concentration; c_{50} , the concentration at which half of that plateau is reached; and n , which sets how steeply $\phi_{\max}$ rises with $\log c$.

Why this function. It follows from a mass balance on the interface. Emulsifying a volume fraction ϕ of oil into droplets of diameter d creates $6\phi/d$ of interface per unit emulsion volume. Covering it at a surface load Γ requires a mass of stabilizer of $6\Gamma\phi/d$, while the stabilizer available is $c \cdot \rho_w (1 - \phi)$, because c is expressed per mass of aqueous phase. Setting the two equal gives $\phi/(1 - \phi) = c/c_{50}$ with $c_{50} = 6\Gamma/(\rho_w d)$, which is the model above with $n = 1$ and $\phi_\infty = 1$. $\phi_\infty < 1$ accounts for the separate ceiling (packing, film drainage or phase inversion near 90 wt%) that no amount of stabilizer removes, and a free exponent n measures how far a stabilizer departs from simple coverage of fixed-size droplets: $n < 1$ means diminishing returns, $n > 1$ means a threshold or cooperative behaviour. The model therefore has a physical basis and turns each curve into three interpretable numbers rather than describing it qualitatively.

What the fit tells us. (i) *Tween 80* follows the coverage model almost exactly ($n = 0.98 \pm 0.08$). Its c_{50} of 0.15 wt%, combined with a typical surfactant load of $\Gamma \approx 2\text{--}3 \text{ mg m}^{-2}$, implies a droplet diameter of $\approx 8\text{--}12 \text{ }\mu\text{m}$, a realistic size that can be checked against microscopy; this supports interface coverage as the factor limiting the oil fraction. (ii) *Native BSA* rises more steeply than coverage alone predicts ($n = 2.1 \pm 0.4$) and only becomes effective above ≈ 3 wt%, consistent with a threshold concentration for forming a cohesive protein film within the emulsification time (slow adsorption and unfolding). (iii) The *PDDA/BSA coacervate* has the lowest c^* (0.16 wt%, versus 0.64 wt% for Tween 80 and 6.5 wt% for native BSA, i.e. $\approx 4\times$ and $\approx 39\times$ less stabilizer for a high-internal-phase emulsion) and a shallow, sub-linear dependence ($n = 0.48 \pm 0.10$). Its fitted c_{50} (0.013 wt%) lies below the lowest concentration measured, so it is an extrapolation; taken at face value it would imply droplets of $\approx 90\text{--}134 \text{ }\mu\text{m}$ for a protein-like surface load. Because that is implausibly large, the coacervate probably does not act through monolayer coverage alone: thick or multilayer interfacial films, or a network in the continuous phase, would explain why so little material stabilizes so much oil. Droplet-size measurements (a limited-coalescence plot of $1/D$ versus stabilizer-to-oil ratio) and data below 0.1 wt% would test this. **Caveats:** 6–8 points per series for 3 parameters, no replicate information, digitized values and wt% used in place of volume fraction. The threshold concentrations c^* are robust (they agree within $\approx 15\%$ with direct interpolation of the data), whereas c_{50} and n should be treated as semi-quantitative.